

HESHALA MADHUWANTHA

GRADUATE [BET(HONS)] IN MECHATRONICS

📍 Pitipana, Homagama 📞 +94 78 939 1396 ✉ heshalamadhuwantha0@gmail.com

🌐 [Heshala Maduwantha](#) 🐙 [Heshala Maduwantha](#)

PROFILE

Interested in achieving a suitable placement in the field of automation Industry and Adept at designing, developing, and optimizing complex mechatronic systems, with a proven ability to drive project success through interdisciplinary collaboration and problem-solving. Seeking to leverage and enhance my skills, knowledge, and experience while contributing to organizational growth and success.

EDUCATIONAL BACKGROUND

Bachelor of Engineering Technology (BET) Honours Degree - Mechatronic Technology

University of Sri Jayewardenepura
2022 - 2026

Bachelor of Information technology (BICT) External Degree - (HDIP Completed)

University of Colombo
2023 - Present

G.C.E. (Advanced Level) Examination - 2020

Information & Communication Technology : A
Engineering Technology : B
Science For Technology : A
Z-Score : 2.7155

G.C.E. (Ordinary Level) Examination - 2017

8 A Passes | 1 C Pass

WORK EXPERIENCE

Engineering Intern

CBL Group | No. 555, High-Level Road, Makumbura, Pannipitiya, Sri Lanka
Oct 2024 - Apr 2025 (06 months)

PROJECTS & PRACTICAL WORK

Design and Development of an Automated Guided Vehicle (AGV) for Indoor Environments

Designed and developed an Automated Guided Vehicle for material handling applications. The system uses embedded control and sensor-based navigation to follow a predefined path, avoid obstacles, and transport loads efficiently. The project focused on automation, motor control, system integration, and improving operational efficiency in industrial environments. ([View Project](#))

Delta Robot Design and Control System Development

Implemented embedded control for a Delta Robot by programming a microcontroller to coordinate multi-axis motor control

Embedded Systems, Automation & Software Projects

Designed and developed electronics, automation, and software-based projects including a USB Type-C to 5V converter with PCB layout, schematic design, and enclosure design using KiCAD and SolidWorks. Developed a number plate detection system for machine vision applications using OpenCV (Pure Image Processing). Also created a washing machine PLC control program using Structured Text, including overload protection and emergency stop functions, and tested it through simulation. ([View Projects](#))

Desktop Inventory Management System for Spare Parts Shop

Developed a Flutter desktop application for a spare parts shop to manage inventory, billing, sales records, and reports. The system includes product add/update/delete, stock tracking, invoice handling, dashboard overview, login, settings, and notifications. SQLite was used for local offline data storage, while Git and GitHub were used for version control and release management.[Flutter, Dart, SQLite, Git, GitHub, Windows Desktop App Development]([View Projects](#)).

CAD Design Projects

Designed and developed multiple industrial and mechanical engineering projects using CAD tools, including a heat exchanger system with P&ID and 3D plant modeling using AutoCAD Plant 3D, quadcopter assemblies, mechanical differential systems, load-cell mounted tank frames, and industrial safety platforms for plant areas using Solidworks. Experienced in 3D CAD modeling, piping and plant layout design, structural frame development, and fabrication-oriented engineering design.([View Projects](#)).

SKILLS

TECHNICAL SKILLS

- Embedded Systems
- Hydraulic & Pneumatic Systems Design
- CAD Designing & Simulation
- Control Systems & PID Controlling
- PLC System Design & Programming
- HMI Designing
- PCB Designing
- Graphical Programming
- Motor Controlling
- C | Arduino | Python Programming

TOOLS

- SolidWork
- Autocad(Electrical & Plant 3D)
- Labview
- Matlab
- Arduino
- EagleCAD
- KiCAD
- Proteus
- Easyada
- Microsoft 365
- TIA portal
- GXworks
- Codesys
- CCW
- Softmaster
- PLstudio
- kinco HMIware
- ROBODK
- EPLAN
- PRIMAVERA P6

GENERAL SKILLS

- Troubleshooting mechanical & automation systems
- Read & Interpret technical drawings
- Hands-on experience with industrial tools and equipment
- Proficient in document management and project planning
- Industrial safety awareness
- Understanding of quality, efficiency improvement, and structured problem-solving practices in industrial environments.
- Programming and Designing knowledge for Automation

CERTIFICATIONS

- **Microcontrollers And Robotics Part Time Training Program**
Ceylon - German Technical Training Institute
- **Information & Communication Technology Certificate Course**
NYSC - National Youth Service Council
- **Python For Beginners(E-certificate course)**
Centre for open & distance Learning(CODL) | University of Moratuwa

REFERENCES

Eng. Peshan Sampath,

Lecturer,

Department of Materials and Mechanical
Technology,

Faculty of Technology,

University of Sri Jayewardenepura

Email : sampathwhp@sjp.ac.lk

Telephone : +94 71 014 9260

Dr. Akila Wijethunga

Senior Lecturer,

Department of Materials and Mechanical
Technology,

Faculty of Technology,

University of Sri Jayewardenepura

Email : akilawijethunga@sjp.ac.lk

Telephone : +94 77 607 8622

I would like to state that all the details supplied here are true and accurate to the best of my knowledge & belief.



Heshala Madhuwantha